

HUMAN ANATOMY • UNIT 1

Core Study Pack

Foundations • Body Chemistry • The Cell • Tissues I & II

Built from the Unit 1 lecture decks (L1-L5) and the Unit 1 lecture exam study guide. Use the companion quiz site ([index.html](#)) to practice one item at a time. Answer tags look like L4 1 A (lecture, item, choice).

Must-know comparisons

Pair	Side A	Side B	Memory hook
Anatomy vs physiology	Structures (internal & external)	How those structures function	Form decides function
Prone vs supine	Face down	Face up	Supine = looking up
Axial vs appendicular	Trunk (head, neck, torso)	Arms and legs	Axis vs appendages
Somatic vs visceral	Superficial: skin, skeletal muscle	Internal organs in cavities	Soma = body wall
Hydrophilic vs hydrophobic	Affinity for water (charged/polar)	No affinity (uncharged)	Philo = love
Solvent vs solute	Does the dissolving (water)	What is dissolved	Water is the solvent
Hypertonic vs hypotonic	More solute outside → cell shrinks	Less solute outside → cell swells	Hyper = extra stuff out
Tight vs desmosome vs gap	Seals; nothing between cells	Strongest weld; not a full seal	Gap = passageway / talk
Hyperplasia vs hypertrophy	More cells	Bigger cells	Plasia = formation
Apoptosis vs necrosis	Programmed; contents stay in	Damage; contents spill + inflammation	Apo = neat suicide
Totipotent vs pluripotent	Any cell type (zygote)	Several types (iPSC / ICM)	Toti = total
Proto-oncogene vs tumor-suppressor	Go signal (healthy)	Stop signal (healthy)	Mutated go = oncogene
Simple vs stratified epithelium	1 layer; weak barrier; does organ work	Many layers; strong barrier	Simple = exchange
Mucous vs serous membrane	Cavities open to exterior	Cavities closed to exterior	Mucus faces the world
Skeletal vs cardiac vs smooth	Voluntary, striated, on bones	Involuntary, striated, heart	Smooth: vessels + gut, no stripes

Macromolecule	Job	Must-know subtypes
Carbohydrates	Energy, metabolism, structure	Glucose (ATP), glycogen (animal store), starch

		(plant store), cellulose (plant wall)
Lipids	Energy, membranes, hormones	Triglycerides (diet/energy), phospholipids (bilayer), steroids (4 rings)
Proteins	Cellular function	Enzymes, insulin, transporters, cytoskeleton
Nucleic acids	Instructions for proteins	DNA → mRNA → protein

L1 • Foundations of anatomy

Core definitions

Definition. Anatomy is the study of internal and external structures.

Example. Heart chambers, femur landmarks, regional terms such as antecubital.

Why it matters. You cannot interpret imaging, surgery, or symptoms without a shared map of the body.

Definition. Physiology is the study of how anatomical structures function. Structure determines function.

Example. Heart next to lungs so blood can pick up oxygen.

Why it matters. Explains why a structural defect changes performance.

Definition. Congenital means a disease or condition present from birth.

Example. Situs inversus: left-right organ reversal; front-back stays normal.

Why it matters. Congenital ≠ always genetic; it only means present at birth.

Definition. Homeostasis is maintaining a stable internal environment.

Example. Blood glucose, body temperature, fluid balance.

Why it matters. Every organ system exists, in part, to defend this stability.

Definition. Supine is face up; prone is face down.

Example. Supine regional-term diagrams vs prone diagrams in lecture.

Why it matters. Patient positioning language is used every day in clinic and lab.

Directional terms, planes, organization

Superior / inferior = above / below. Anterior / posterior = front / back. Medial / lateral = toward / away from midline. Proximal / distal = closer to / farther from the trunk attachment. Superficial / deep = toward surface / farther in.

Planes: frontal (coronal) = anterior vs posterior; sagittal = right vs left; transverse = superior vs inferior. CT scans evaluate transverse slices.

Levels (small → large): chemical (atoms, molecules, electrolytes) → cell (smallest living unit) → tissue → organ → organ system → organism. Electrolytes (Na⁺, K⁺) and organic molecules are obtained from food and are not living.

Axial body = trunk. Appendicular body = arms and legs. Somatic body = superficial structures (skin, skeletal muscle). Visceral body = internal organs in cavities.

Quadrants from the lecture slide: RUQ gallbladder, LUQ stomach, LLQ colon, RLQ appendix.

Regional terms (memorize)

Abdominal (belly), antebrachial (forearm), antecubital (front of elbow), axillary (armpit), brachial (arm), buccal (cheek), carpal (wrist), cephalic (head), cervical (neck), coxal (hip), crural (leg), femoral (thigh), frontal (forehead), inguinal (groin), manus (hand), mental (chin), nasal (nose), oral (mouth), palmar (palm), patellar (kneecap), pedal (foot), pelvic (pelvis), plantar (sole), pubic (pubis), sternal (sternum), tarsal (ankle), thoracic (chest), umbilical (navel).

Eleven organ systems — lecture “red text” jobs

Integumentary: external covering; protects deeper tissue; sweat/oil glands. Skeletal: protects organs; framework for muscle; blood cells in bone; stores minerals. Nervous: fast-acting control; electrical signals. Muscular: conscious movement, posture, heat. Cardiovascular: heart pumps; vessels transport; blood carries O₂, nutrients, hormones, wastes. Endocrine: slow-acting control; glands secrete hormones. Respiratory: supply O₂, remove CO₂. Lymphatic: return leaked fluid; immunity. Digestive: break food into absorbable nutrients. Urinary: eliminate liquid waste; water / electrolyte / acid–base balance. Reproductive: production of offspring.

L2 • Body chemistry

Definition. Water is a versatile solvent because of its charge (polarity), so it attracts other charged molecules.

Example. *NaCl dissolving into Na⁺ and Cl⁻.*

Why it matters. Almost every transport and reaction in the body happens in water.

Definition. Osmosis is diffusion of water across a membrane, toward the side with more solute.

Example. *Hypertonic seawater pulls water out of cells. Hypotonic water intoxication swells cells.*

Why it matters. IV fluids are isotonic so cells neither shrink nor burst.

Plasma membrane jobs: barrier + selective traffic. Most of the membrane is a phospholipid bilayer. Hydrophobic molecules cross easily; hydrophilic molecules and ions need proteins. Active transport moves solute against its gradient and requires ATP.

Bulk transport (endocytosis): phagocytosis = cellular eating (solids); pinocytosis = cellular drinking (fluid); receptor-mediated endocytosis = specific cargo (example: cholesterol).

Newborns have limited glycogen in liver and skeletal muscle, so they are prone to hypoglycemia.

L3 • The cell and cell division

Junctions: tight = complete seal, nothing passes between cells; desmosomes = strongest mechanical link, not a complete fusion; gap = passageway and communication.

Cell change: hyperplasia = more cells; hypertrophy = bigger cells; atrophy = fewer or smaller cells.

Death: apoptosis = programmed, contents digested inside, no spill; necrosis = after severe damage, contents spill, inflammation.

Three cell parts: plasma membrane, cytoplasm/cytosol (intracellular fluid), nucleus.

Organelle jobs (lecture framing): mitochondria = efficient ATP because of folded membranes; RER = proteins for export / membranes / lysosomes; free ribosomes = proteins for the cell; Golgi = modify/pack (adds carbohydrate); lysosomes = digest; peroxisomes = detox / “immune-like”; nucleus = DNA; nucleolus = makes ribosomes; centrosomes + microtubules = mitotic spindle; cytoskeleton = support and movement; enzymes = speed reactions.

Cytoskeleton sizes: microfilaments (smallest; microvilli) < intermediate filaments < microtubules (largest; spindle, cilia, flagella). Variations: cilia beat; flagella give directional movement; microvilli increase surface area; spindle participates in division.

DNA + histones = chromatin/chromosomes. Histones package DNA so it fits and can be regulated. Transcription makes mRNA; translation makes protein. Flow: DNA → mRNA → protein. Genetic disease (sickle cell) starts as faulty DNA.

Cell cycle: interphase (most of life: G1, S = DNA copy / sister chromatids form, G2) and mitosis. Homologous chromosomes = non-identical pair from parents, always present. Sister chromatids = identical copies that exist after S and are pulled apart in anaphase.

Mitosis: prophase = spindle forms and nuclear membrane breaks down; metaphase = chromosomes at equator; anaphase = sisters pulled apart; telophase = two nuclei form.

L4 • Embryology and epithelium

Zygote = fertilized egg. Stem cell = unspecialized cell that can make more cells. Totipotent = any cell type. Pluripotent = several cell types. iPSC = induced pluripotent stem cell (adult cell reprogrammed). Differentiation = immature cells take on specialized form and function. Gastrulation = embryo reorganized into multilayered germ layers. Neurulation = neural tube formation. Teratogen = agent that causes congenital defects (most dangerous during embryonic organogenesis).

Prenatal stages: pre-embryonic (shortest), embryonic (organogenesis), fetal (longest).

Germ layers form at gastrulation. Ectoderm (outer) → nervous system + epidermis. Mesoderm (middle) → somatic body, CV system, muscle, bone, CT. Endoderm (inner) → lining of digestive, respiratory, reproductive tubes.

Cell-cycle genes: proto-oncogenes stimulate division (growth proteins). Tumor-suppressor genes inhibit division (brakes, plus membrane proteins that sense crowding / attach cells). Mutated proto-oncogenes = oncogenes (too much go). Failed suppressors = cancer because the brakes are gone.

Epithelium functions: barrier + organ activity (secrete, absorb, filter). Features: packed cells, polarity, basement membrane, avascular but highly mitotic (heals well). Simple = one layer, poor barrier, good at organ work (air sacs, gut, kidney). Stratified = many layers, good barrier, less exchange (skin, mouth). Shapes: squamous flat, cuboidal cube, columnar tall, pseudostratified looks layered but is simple, transitional stretches. Accessories: cilia on pseudostratified columnar; microvilli on simple columnar; keratin on stratified squamous.

L5 • Tissues part II

Four tissues: epithelial (barrier + organ function), connective (support; least cellular), muscle (movement), nervous (electrical signals).

Three membranes: cutaneous (skin), mucous (open to exterior), serous (closed cavities). Membranes are usually epithelial (plus connective support). Mesothelioma = cancer of serous membrane cells.

Connective tissue: cells + protein fibers + ground substance (fibers + ground substance = matrix). Resident cells physically support and stay put; wandering immune cells come from blood. Fibers: collagen (strong, nearly universal), elastic (stretch), reticular (mesh), plus related structural proteins. Classifications include loose (areolar, adipose, reticular), dense, cartilage, bone, and fluid (blood).

Muscle: skeletal = voluntary, striated, on bones; cardiac = involuntary, striated, heart; smooth = involuntary, no striations, vessels and digestive tract.

Nervous tissue cells: neurons and glia. Blood and tendons/ligaments are connective tissue.

Vocabulary list

Anatomy, physiology, congenital, homeostasis, prone, supine, superior, inferior, anterior, posterior, medial, lateral, proximal, distal, superficial, deep, frontal plane, sagittal plane, transverse plane, axial, appendicular, somatic, visceral, electrolyte, hydrophilic, hydrophobic, solvent, solute, osmosis, isotonic, hypotonic, hypertonic, hypoglycemia, glucose, glycogen, starch, cellulose, triglyceride, phospholipid, steroid, phagocytosis, pinocytosis, receptor-mediated endocytosis, tight junction, desmosome, gap junction, hyperplasia, hypertrophy, atrophy, apoptosis, necrosis, cytosol, mitochondrion, Golgi, lysosome, peroxisome, ribosome, RER, SER, nucleolus, centrosome, microfilament, microtubule, intermediate filament, histone, chromosome, transcription, translation, interphase, mitosis, homologous chromosome, sister chromatid, prophase, metaphase, anaphase, telophase, zygote, stem cell, totipotent, pluripotent, iPSC, differentiation, gastrulation, neurulation, teratogen, ectoderm, mesoderm, endoderm, proto-oncogene, oncogene, tumor-suppressor gene, simple epithelium, stratified epithelium, squamous, cuboidal, columnar, pseudostratified, transitional, cilia, microvilli, keratin, mucous membrane, serous membrane, cutaneous membrane, collagen, elastic fiber, reticular fiber, ground substance, skeletal muscle, cardiac muscle, smooth muscle, neuron, glial cell.

Quiz bank + answer key

The live quiz bank lives in the website files (questions.js). Each item is tagged by lecture (example L4 1 A). Types: multiple choice, true/false, short answer, and application. Correct answers can be excluded on later runs.

How to use: open index.html → pick a lecture → answer one item → read the explanation → later sessions skip mastered IDs until you reset progress.

Spot-check key from the decks:

L1 10 C CT plane = transverse. L1 18 C Appendix = RLQ. L1 20 B Fast control = nervous.

L2 9 C ATP carb = glucose. L2 13 B Amphipathic lipid = phospholipid. L2 18 B Seawater is hypertonic.

L3 1 C Tight junction seals. L3 6 B Apoptosis is programmed. L3 22 B Metaphase = equator.

L4 2 B Totipotent = any cell. L4 10 C Nervous system = ectoderm. L4 19 B Cilia = pseudostratified columnar.

L5 5 B Open cavities = mucous. L5 10 C Conscious muscle = skeletal. L5 15 A Blood is connective tissue.